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INCH-POUND

A-A-59929
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COMMERCIAL ITEM DESCRIPTION

Generator, Signal

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for PD13WRGUEEA12 for all federal agencies.

1. SCOPE

1.1 Scope. This Signal Generator generates a highly stable continuous wave and modulated signal. The unit is capable of generating Amplitude Modulation (AM), Frequency Modulation (FM), Phase Modulation (ΦM) and Pulse Modulation (PM) with internal and external signals for modulation. This instrument provides signals for, the development, calibration, repair and maintenance of navigational aids, satellite and other communications equipment.

2. CLASSIFICATION

2.1 Classification. The test set covered by this CID shall be commercially available equipment that has factory installed options and/or may be modified to the extent necessary to meet the following description. The test set shall be Class 3, in accordance with MIL-PRF-28800. Class 3 equipment is designed for operational environments where conditions are routinely encountered in a diverse, controlled climate. Examples include test equipment for bench-top, rack-mount and occasional portable use in diverse, environmentally controlled locations.

3. SALIENT CHARACTERISTICS

3.1. Item Description. Signal Generator for Microwave Radio Frequency.

3.2. Radio Frequency Characteristics. The requirement for the Signal Generator is to generate a signal with a span from 100 kHz to 2.5 GHz. The signal generator shall have an RF sweep

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function. This signal generator shall provide RF output with AM, FM, PM, ΦM and have external modulation capability. The signal generator shall have external pulsed carrier function capability. The signal generator shall have at least 8 marker frequencies.

3.3. Radio Frequency Resolution. The signal generator frequency resolution shall be better than 0.001 Hz in in constant wave (CW) with a 10 MHz internal frequency reference.

3.4. Concurrent Modulation. The signal generator shall be capable of generating an output signal with simultaneous FM, AM and Phase modulation content in pulse modulated intervals.

3.5. Pulse Modulation. The signal generator shall be capable of providing pulse modulation durations equal to or less than 1 microsecond (≤1μS) with a repetition frequency equal or greater than 50 kHz.

3.6. RF Spectral Purity. The signal generator shall have a minimum of ≤60dB and ≤90dB average harmonic noise content throughout the spectrum while under modulation.

3.7. Sweep Functions. The signal generator shall be capable of performing linear, logarithmic, continuous step frequency sweeps and list sweep in less than 100 seconds and with less than ±10 dB variation in output level.

3.8. Marker Frequencies. The signal generator shall have the capability to provide at least two marker frequencies (points). These can be entered manually through the user menu or the GPIB.

3.9. Frequency Accuracy. The frequency accuracy internal reference frequency aging shall be $\pm 3.0X10^{-8}$ /Year or less than ±0.5 % at maximum sweep time, see table 1.

Table 1 Spectral Purity Content from Center Frequency F_0 (dBc/Hz)

	Frequency Range (MHz)					
	0.01 to 119.9	120 to 159.9	160 to 319.9	320 to 639.9	640 to 1279.9	1280 to 3000
	Content in dBc/Hz	Content in dBc/Hz	Content in dBc/Hz	Content in dBc/Hz	Content in dBc/Hz	Content in dBc/Hz
1 Hz	≤65	≤66	≤60	≤54	≤48	≤42
10 Hz	≤95	≤96	≤90	≤84	≤78	≤72
100 Hz	≤110	≤115	≤109	≤103	≤97	≤92
1 kHz	≤120	≤129	≤124	≤118	≤112	≤106
3 kHz	≤120	≤129	≤124	≤118	≤112	≤106
5 kHz	≤125	≤131	≤126	≤120	≤114	≤108
10 kHz	≤130	≤142	≤136	≤131	≤124	≤118
100kHz	≤130	≤142	≤136	≤131	≤124	≤118

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3.10. Spurious Signals. The total spurious signal content, including side bands & sub harmonics, accompanying the desired RF signal shall be at least 40 dB below the output level of the center frequency signal at maximum modulation, see table 2.

Table 2 Spurious Signal Content Levels

	Carrier frequency range (MHz)					
Spurious signals	0.1 to 120	120 to 160	160 to 320	320 to 640	640 to 1280	1280 to 3000
Spurious non-harmonically related.2	<-90 dBc	<-100 dBc	<-96 dBc	<-90 dBc	<-84 dBc	<-78 dBc
Sub-harmonically related (f/2, 3f/2, etc.).	none	none	none	none	<-70 dBc	<-40 dBc
Power line (60 Hz) related or microphonically generated (within 300 Hz).	<-90 dBc	<-85 dBc	<-80 dBc	<-75 dBc	<-70 dBc	<-65 dBc
Harmonics	<-30 dBc, ≤+13 dBm output <-25 dBc, +13 dBm to +16 dBm output					<-25 dBc
	Frequency range (MHz)					
	Front Panel					**Rear Panel**
Offset from carrier	0.01 to 119.9	120 to 159.9	160 to 319.9	320 to 639.9	640 to 1279.9	1280 to 2559.9	640
1 Hz	-76	-74	-68	-62	-56	-50	-62
10 Hz	-106	-104	-98	-92	-86	-80	-92
100 Hz	-124	-123	-117	-111	-105	-100	-112
1 kHz	-126	-129	-124	-118	-112	-106	-121
3 kHz	-126	-129	-124	-118	-112	-106	-121
5 kHz	-128	-131	-126	-120	-114	-108	-129
10 kHz	-132	-142	-136	-131	-124	-118	-145
100 kHz	-132	-142	-136	-131	-124	-118	-157

3.11. Modulation. The signal generator shall be capable of generating AM, FM, ΦM, with internal and external signals and PM with external signals for modulation.

3.12. Harmonic Distortion. The total harmonic distortion of the modulated signal shall be less than 5.0% for amplitude, frequency, and phase modulation except when Gaussian (White or Pink Noise) signal is applied.

3.13. Power Output Level. The signal generator shall be able of providing a minimum of +16 dB/mW @ 50 Ω (ohms) throughout the spectrum with filters enabled (on). The Signal Generator shall have configuration options to include attenuators and/or higher power output to have a minimum span of –115 dB to +16 dB throughout the spectrum.

3.14. Broadband & Single Side Band Noise. The level of any broadband noise on the desired RF signal shall be at ≤100 dB below the carrier output level, as measured at the RF OUT connector with 0 dB/mW @ 50 Ω load, see table 3.

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Table 3 Single Side Band Noise Average Weight

	Frequency range (MHz)					
	0.01 to 119.9	120 to 159.9	160 to 319.92	320 to 639.9	640 to 1279.9	1280 to 3000
	Residual Noise Range dB/mW @ 50 Ω with Single Side Band Modulation					
10 Hz	-108 ~ -114	-112 ~ -119	-106 ~ -113	-100 ~ -107	-93 ~ -101	-88 ~ -95
100 Hz	-121 ~ -126	-122 ~ -129	-118 ~ -124	-112 ~ -119	-105 ~ -112	-100 ~ -106
1 kHz	-128 ~ -133	-131 ~ -138	-127 ~ -134	-121 ~ -128	-115 ~ -122	-109 ~ -116
3 kHz	-128 ~ -136	-131 ~ -139	-127 ~ -135	-121 ~ -129	-115 ~ -123	-109 ~ -117
5 kHz	-129 ~ -138	-133 ~ -141	-129 ~ -136	-123 ~ -130	-117 ~ -124	-111 ~ -118
10 kHz	-132 ~ -137	-142 ~ -147	-136 ~ -142	-131 ~ -136	-124 ~ -130	-118 ~ -124
100kHz	-132 ~ -137	-142 ~ -147	-136 ~ -142	-131 ~ -136	-124 ~ -130	-118 ~ -124

3.15. RF Output Attenuation. The RF level shall be adjustable, in not greater than 1 dB increments, from 0 to -120 dB, when measured into a 50 Ω load at the RF output connector. Output flatness shall be ±1dB average throughout usable spectrum.

3.16. Noise Figures. The broadband noise figures (signal to noise ratio in dB) shall be ≥30 dB average throughout the spectrum at 0 dB/mW @ 50 Ω.

3.17. Triggering. The signal generator shall have external triggering inputs that trigger the sweep function for Analog Frequency, Step Frequency, and CW Power Sweep functions. Triggering voltages shall be TTL level user switchable HI or LOW. Sweep control shall be an internal menu driven and the external modulation must be controlled thru a variable proportional voltage source (typical 0~10 V_{DC}).

3.18. Internal and External Reference. The generator shall have an external reference input and output connector for calibration and comparison procedures.

3.19. Design and Construction. The signal generator shall be used in both bench top and rack mount environments. The Signal Generator is a commercial of the shelf item subject to commercial products regulatory standards. The Signal Generator shall comply with all applicable MIL-PRF-28800 sections.

3.20. System Safety. The signal generator shall comply with all sections of the Underwriters Laboratories (UL) standard 60950 for electrical safety. The system shall comply with MIL-STD-1472 section 5.9.11.3.3 for weight & size restrictions for single person lift & handling.

3.21. Sound Levels. The signal generator acoustic noise shall conform to sound level requirements on section 1 of ISO 7779 measured per section 2 of ISO 8253.

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3.22. Electrostatic Discharge (ESD). The design of the signal generator shall preclude equipment damage due to ESD, protect personnel from electrical shock due to static charging, and prevent ignition of explosive atmospheres due to sparking in accordance with MIL-STD-1686 section 5.2.2.3.

3.23. Electro Magnetic Interference (EMI). The signal generator EMI emissions shall be in accordance with the following radiated emission and susceptibility requirements of MIL-PRF-28800, MIL-STD-461: RE102 and RS103 & IEC-1000-4 thru IEC-1000-6. Emitted radiation absorption and radiation limits FCC Part 15.109 & 15.209.

3.24. Service Life. The design of the signal generator shall be such that under normal use and operation the service repair can be performed according to MIL-PRF-28800 sections 3.14.2 & 3.14.3.

3.25. Port Damage. The output port shall withstand up to 5.0 VDC or VAC_{RMS} or 27dBm (typical) levels without damaging the internal circuitry. Protection can be achieved by either internal or external fusing.

3.26. Connectors. The system shall have an “N” type connector, for radio frequency output. BNC type connectors for other front and rear panel inputs and outputs. A General Purpose Interface Bus (GPIB) interface with IEC-60488-2 Protocol connector is required for compatibility and control functions. Connectors shall meet dimensions and form as shown in sections 16H through 24D of MIL-PRF-39012

3.27. Operating Temperature Range. The signal generator shall meet the specifications listed herein with full capabilities within the temperature range of 32° to 122 °F (0 °C to 50 °C) after tests in MIL-PRF-28800 section 4.5.5.1.1 have been performed for class 3 equipment classification (Per table 11 on page 41).

3.28. Storage Temperature Range. The signal generator shall meet and/or exceed manufacturers specifications for stored equipment after tests of MIL-PRF-28800 section 4.5.5.1.1 temperature, humidity & storage have been performed for instruments in the range of class 3 equipment classification (Per table 11 on page 41).

3.29. Altitude Requirements. The signal generator shall meet its operating performance and accuracy while operating at an altitude of 15,090 ft. (4,600 meters). The equipment shall meet its performance and accuracy requirements, for operation and non-operating storage, as specified in MIL-PRF-28800 section 3.8.3 and section 4.5.5.2 respectively at this altitude.

3.30. Calibration & Maintenance Adjustments. No special tools, proprietary software or devices shall be required for alignment, adjustment or calibration except those furnished with the test set.

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3.31. Calibration Period. The design of the signal generator shall provide readily accessible calibration and maintenance adjustments. The calibration adjustments shall be accessible without removal of the test set case or modules. The calibration interval shall be based on a period of one calendar year based on an operating time of 8,776 hours. At the end of this interval 85% of the units shall remain within calibration tolerances in accordance with MIL-PRF-28800 section 3.12.

3.32. Reliability. The design of the equipment shall be such that under normal use and operation the mean time before failure (MTBF) exceeds 1,500 hours of operation accordance with MIL-PRF-28800 section 3.13. The manufacturer shall provide a summary of maintenance or warranty records to establish the validity of the statistical assertion in accordance with MIL-PRF-28800 section 3.13. The manufacturer shall be able to reasonably establish the claim that the offered equipment is capable of the required level of reliability by meeting and/or exceeding MIL-PRF-28800 sections 3.13 and section 4.5.8.1.

3.33. Self-Test Capability. The signal generator shall be capable of providing self-diagnostics during the power on cycle or on demand by user input request. The displayed information provided shall indicate whether the instrument is operating within the performance specifications, specific conditions and time signatures of events relevant to the calibration and general condition of the instrument in accordance with section 3.3.6 of MIL-PRF-28800.

3.34. Maintainability. The signal generator shall meet the human engineering factors of ergonomics for operational & maintainability criteria, providing access to controls, internal components and adjustments in accordance with MIL-STD-1472 paragraphs 5.9 through 5.9.18 and MIL-PRF-28800 section 6.6.2.4.

3.35. Preventive Maintenance. Preventive maintenance shall not require more than 15 minutes per a 30-day period. Preventive maintenance shall not require breaking of the equipment seams where calibration seals would normally be placed in accordance with MIL-PRF-28800 section 3.14.2.

3.36. Corrective Maintenance. The signal generator shall be designed to meet reparability criterion in accordance with section 3.14.1 of MIL-PRF-28800.

3.37. Electrical Power Sources & Connections. The equipment shall operate from a nominal voltage of 120 V_{RMS} commercial or military single phase AC (±10%) at line frequencies of 50Hz, 60Hz, in accordance with MIL-PRF-28800 paragraph 3.1.5.2 and 3.1.5.2a. Transient state conditions shall meet MIL-PRF-28800 requirements for the classification specified herein. AC input power shall be used to power the Signal Generator and recharge batteries if equipped with them.

3.38. Electrical Surge. The signal generator shall be able to provide surge protection that meets section 38 of the UL 1449 standard for equipment and personnel protection.

3.39. Noise Suppression. The signal generator shall be able to operate with electrical noises and harmonic content present on the alternating current power source with frequencies in the 30Hz to 50 kHz range at 41.6 dBV ±1 dBV (108 VAC to 132 VAC) level measured at the power input in accordance with UL 1283, IEC 60939 and under the guidance of the IEEE-519 standard.

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3.40. Dimensions. The signal generator shall comply with width and depth of EIA/ECA-310-E standard equipment rack configuration. The height shall be compatible with two (2U) to four standard height units (4U). One unit or "U" is defined by EIA/ECA-310-E to be 1.75 inches or 44.45 mm.

3.41. Weight. The total weight of the signal generator, excluding accessories and manuals, shall not exceed 75 pounds, (34 kilograms).

3.42. Rack Mount. The signal generator shall be capable of being rack-mounted in a standard EIA/ECA-310-E equipment rack and include a rack-mount conversion kit if needed.

4. REGULATORY REQUIREMENTS

4.1 Recycled recovered materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

5. PRODUCT CONFORMANCE PROVISIONS

5.1. Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and the product conforms to the producer’s engineering drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract

6. PACKAGING.

Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

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7.1 Source of documents.

7.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained at https://assist.dla.mil/ or available from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094

7.1.2 ANSI/EIA documents can be obtained online at http://www.eciaonline.org/eiastandards or TechAmerica, 601 Pennsylvania Avenue, NW North Building, Suite 600, Washington, DC 20004.

7.1.3 IEC documents can be obtained online at http://www.iec.ch/ or IEC Regional Centre for North America, 446 Main Street, 16th Floor, Worcester, MA 01608.

7.1.4 ISO documents can be obtained online at http://www.iso.org/ or American National Standards Institute, 1889 L St. NW, Washington, DC 20036.

7.1.5 IEEE documents can be obtained online at http://standards.ieee.org/ or IEEE Operations Center, Hoes Lane, 3rd Floor, Piscataway, NJ 08855.

7.1.6 UL documents can be obtained at http://www.ul.com/global/eng/pages/.

7.1.7 FCC documents can be obtained at http://www.fcc.gov/ or Federal Communications Commission, 445 12th Street, SW, Washington, DC 20554

7.1.8 FAR the FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh PA 15250-7954. Electronic copies of the FAR may be obtained from https://www.acquisition.gov/far/index.html

7.2 Key Words

Frequency
Calibration
Modulation

Custodians:
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Preparing Activity:
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Reviewers:
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